

Final Year Project Proposal



Department of Computer Science
Bahria University, Islamabad

RupiSafe: Your Daily Expense Guardian

Supervisor: Dr. Hafiz Ishfaq Ahmad

Submitted by:

01-134222-042 Fabiha Adnan

01-134222-080 Minal Tahir

Abstract

RupiSafe is a personal finance management system designed to tackle the growing financial challenges faced by individuals in many parts of the world. Rising inflation, increasing living costs, and limited financial awareness often lead to overspending and end-of-month financial stress. While several budgeting applications exist, most lack localization for users and provide minimal real-time guidance. RupiSafe offers an AI-assisted budgeting solution that monitors user spending, predicts financial risks, and provides adaptive, data-driven recommendations.

The system features two modes: a standard Money Guide mode for routine budgeting and a Survival Mode that activates once 80% of the user's income has been spent. Survival Mode calculates a daily spending allowance, suggests low-cost alternatives, and delivers real-time advisory support to help users sustain themselves until the next income cycle. The system's AI and NLP components classify expenses, analyze patterns, and generate personalized financial insights.

Since a comprehensive Pakistan-specific dataset is currently unavailable, the project utilizes a foreign dataset for model training and evaluation, with the design structured such that the developed models can later be adapted to Pakistan's context when suitable local data becomes available. Through visual reports, alerts, and trend forecasts, RupiSafe aims to promote informed decision-making, reduce financial stress, and cultivate healthier spending habits.

Contents

1	Introduction	3
2	Objectives	4
3	Problem Description	5
4	Proposed Solution	6
4.1	Money Guide	6
4.2	Survival Mode	6
5	Methodology	8
5.1	User Budget Setup	8
5.2	AI-Based Expense Classification	8
5.3	Spending Pattern Analysis	8
5.4	Survival Mode Activation	9
5.5	AI-Driven Insights	9
5.6	Reporting and Visualization	9
5.7	System Architecture	11
5.8	Flowchart	12
6	Project Scope	13
6.1	In Scope	13
6.2	Out of Scope	13
7	Tools and Technologies	14
7.1	Frontend	14

7.2	Backend	14
7.3	Database	14
7.4	AI/ML	14
7.5	Visualization	15
7.6	Deployment	15
7.7	Data Handling	15
8	Machine Learning and NLP	16
8.1	Expense Categorization (NLP)	16
8.2	Overspending Prediction (ML)	16
8.3	Personalized Recommendations (ML)	16
9	Feasibility Study	17
9.1	Risks Involved	17
9.2	Risk Mitigation	17
9.3	Resource Requirement	17
10	Application Areas	18
11	Expertise of Team Members	19
12	Milestones	20
13	Expected Outcomes	22
14	Significance of Project	23

1 Introduction

In today’s world, managing personal finances is a challenge for many individuals, especially considering how inflation, transport costs, and everyday living expenses continue to rise [1]. The constant increase in utility bills, fuel prices, and essentials has reduced the purchasing power of an average citizen. As a result, families and individuals often struggle to balance their income with their daily expenses. Some do not track their expenses at all.

Students, employees, and families often create a budget at the start of the month, but due to poor tracking, impulsive spending, and lack of financial awareness, they end up overspending [2]. This leads to financial stress and loan dependency by the end of the month. Despite the growing need for structured personal finance management, the tools available to users in certain countries remain limited in both functionality and localization.

Existing budgeting applications such as Mint and YNAB provide basic features, but they cater to a Western audience and do not offer localized guidance. Most importantly, they do not help users manage [3]. Their features focus on visualization, not adaptive decision support.

RupiSafe aims to address these gaps. It provides AI-assisted budgeting with real-time guidance and a unique Survival Mode feature. To achieve this, it uses generalized spending categories like Food, Transport etc, adapted from a combined dataset and refined over time based on user inputs. This approach allows RupiSafe to learn personal spending patterns and offer tailored recommendations, even in the absence of comprehensive Pakistan-specific datasets or real-time APIs from services like Indrive, Yango, or Foodpanda.

2 Objectives

The primary objectives of RupiSafe include:

1. To design a Smart Advisor that monitors spending speed and provides timely warnings.
2. To implement a unique Survival Mode that activates during low funds, calculating daily allowances and suggesting practical low-cost options.
3. To deliver analytics and reports that help users understand their financial behavior and improve long-term habits.

3 Problem Description

Users in Pakistan lack an adaptive, intelligent budgeting tool that not only tracks expenses but also actively guides them during financial shortages to prevent overspending and end-of-month crises.

4 Proposed Solution

RupiSafe functions as a decision support system with two main modes:

4.1 Money Guide

This mode provides:

- Budget setup by categories such as Food, Transport, Rent, Shopping, Misc.
- Smart Advisor alerts for overspending.
- Visualizations including pie charts and progress bars.
- What-If Simulator for predicting the impact of potential purchases.

4.2 Survival Mode

This mode activates when **80% of income is spent**. It provides:

- Daily allowance calculation:

$$A_{survival} = \frac{Remaining\ Budget}{Days\ Left}$$

- Crisis recommendations such as low-cost alternatives.
- Motivational reminders.

Example: Consider a student who has Rs. 3000 remaining with 10 days until payday and a daily spending that surpasses a pattern of Rs 400 on food, Rs 200 on transport and Rs 150 on miscellaneous items. Rupisafe's Survival Mode addresses this by suggesting reductions in spending such as cutting food costs, minimizing transport expenses and pausing non-essential purchases which ensures that the remaining budget can sustain the user until the next income cycle.

5 Methodology

5.1 User Budget Setup

The process begins when users enter their monthly income; they define spending categories like Food, Transport, Rent, Shopping, Miscellaneous etc., and set personal budget limits. The system will classify and track transactions using natural language processing NLP. When expenses exceed **80%** of the income the system automatically activates survival mode.

5.2 AI-Based Expense Classification

Every new transaction added by the user is processed through an NLP based classification module. Transaction descriptions are converted into vector embedding, TF-IDF or pre-trained embedding, and a lightweight model such as Logistic Regression assigns the expense to its relevant category. This enables accurate tracking and category wise analysis.

5.3 Spending Pattern Analysis

- **Time series trend analysis:** Predicts when funds may run out based on spending speed.
- **Benchmark comparison:** Compares the user's category-wise spending with typical local averages, such as the average student food cost.
- **Smart Advisor alerts:** Issues warnings when spending speed exceeds healthy limits.

5.4 Survival Mode Activation

When expenses reach the predefined threshold, typically 80% of income, the system automatically switches into **Survival Mode**.

Survival Mode performs:

- **Daily allowance calculation:**

$$A_{\text{survival}} = \frac{\text{Remaining Budget}}{\text{Days Left}}$$

- **Crisis-focused recommendations:** low-cost alternatives for food or transport.
- **Motivational alerts:** to help the user stay on track.

This mode ensures users can sustain themselves until the next income cycle.

5.5 AI-Driven Insights

The AI engine combines rule-based logic and machine learning predictions to generate personalized guidance, such as:

- Which category needs reduction.
- How much to cut down to stay within limits.
- Spending habits that require improvement.

Clustering techniques, such as K-means, may be applied to identify behavior patterns and refine the advice.

5.6 Reporting and Visualization

A reporting engine compiles:

- Monthly spending summaries
- Category-wise breakdowns
- Smart Advisor alerts
- Survival Mode activity
- Spending trends and forecasts

Visualizations (using Chart.js, D3.js) and downloadable reports (PDF/CSV) help users understand their habits clearly and make informed decisions.

5.7 System Architecture

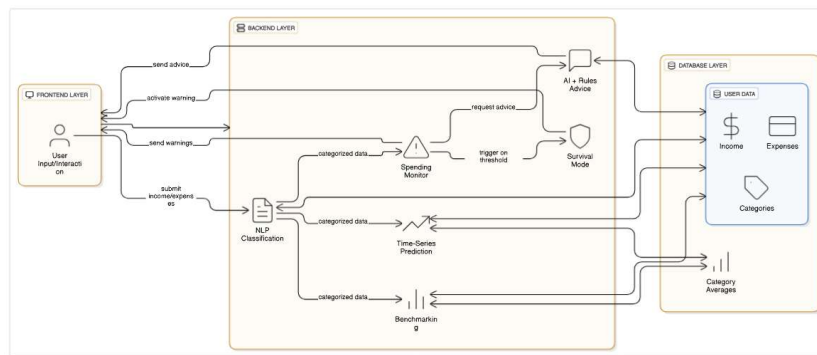


Figure 5.1: Architecture Diagram for RupiSafe

5.8 Flowchart

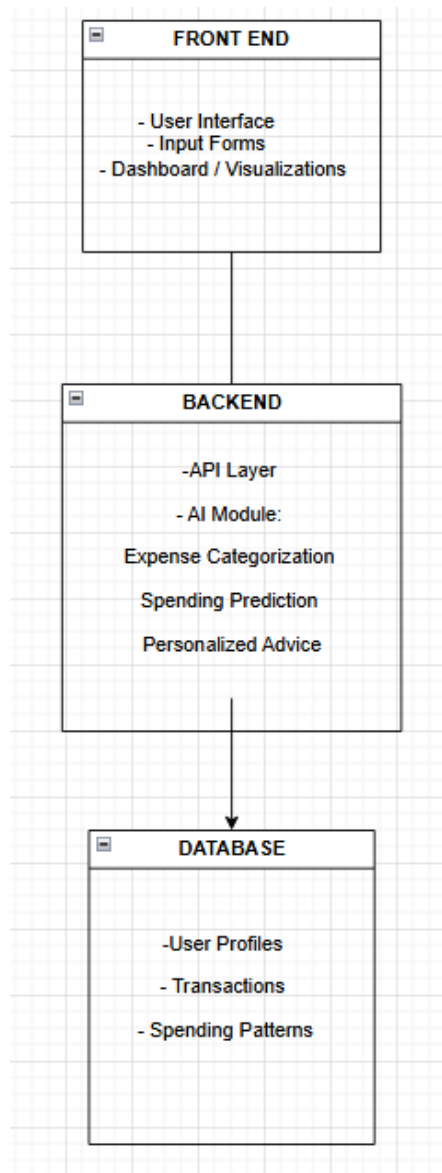


Figure 5.2: RupiSafe Flowchart

6 Project Scope

6.1 In Scope

Mobile interface, expense categorization, visualization, Survival Mode, AI insights, PDF/CSV reports.

6.2 Out of Scope

Bank integration, chatbot, multi-language support, currency conversion.

7 Tools and Technologies

7.1 Frontend

The mobile interface will be developed using Flutter, which allows for cross-platform development, enabling the app to run smoothly on both Android and iOS devices.

7.2 Backend

The backend will be implemented using Node.js to handle real-time updates and manage communication between the frontend and the database. REST APIs will serve as the primary communication layer. Node.js is chosen for its asynchronous capabilities and real-time performance.

7.3 Database

Firebase will be used for real-time data syncing and authentication services, providing seamless updates across devices.

7.4 AI/ML

Python will be used for implementing AI and machine learning features due to its rich ecosystem of libraries. scikit-learn will be employed for regression models to predict overspending trends. NLP libraries like NLTK and spaCy will handle text preprocessing and classification of transaction de-

scriptions. Pandas and NumPy will support data cleaning, feature extraction, and mathematical calculations, including budget averages and Survival Mode computations.

7.5 Visualization

Chart.js and D3.js will be used to create interactive, dynamic visualizations within the mobile app, such as budget progress bars, spending trends, and category breakdowns. For backend Matplotlib and Seaborn will provide static visualizations to support reports and AI insights.

7.6 Deployment

The application will be deployed on platforms such as Firebase and GitHub. Firebase allows for easy hosting and real-time database integration. GitHub will be used for version control, enabling collaborative development, code management, and continuous integration.

7.7 Data Handling

Since real-time APIs from services like Indrive, Yango, or Foodpanda are unavailable and no comprehensive Pakistan-specific financial dataset exists, the system relies on generalized spending categories such as Food (Grocery, Dining, Delivery), Transport (Public, Private), Rent, Shopping, Utilities, and Miscellaneous. The dataset collected for this project provides the initial structure for classifying expenses and identifying spending trends. As users enter their expenses, these inputs allow the system to learn personal spending patterns and refine its recommendations over time. This adaptive approach keeps the system functional and accurate in data-limited environments, with the flexibility to be fully localized for Pakistan when suitable regional data becomes available.

8 Machine Learning and NLP

8.1 Expense Categorization (NLP)

For expense categorization, the system will use natural language processing techniques such as token matching, fuzzy matching, and keyword-based classification, implemented using Python NLP libraries.

8.2 Overspending Prediction (ML)

Overspending prediction will estimate the number of days until a user's money runs out by calculating the remaining budget relative to spending speed. Linear regression models can also be added to capture trend-based predictions.

8.3 Personalized Recommendations (ML)

Personalized recommendations will be generated by learning the user's category averages over time. The system can provide actionable advice, such as suggesting a reduction in private transport spending by a specific percentage to ensure the user survives until the end of the month. These recommendations will operate at the category level to maintain relevance and precision.

9 Feasibility Study

9.1 Risks Involved

Data accuracy may be a concern, as user-entered expenses could be inconsistent. AI limitations also exist because the accuracy of the model depends on the quality of the dataset and labeling. Timeline risks are present since the integration of machine learning components may take longer than planned.

9.2 Risk Mitigation

To address these risks, dummy datasets and test cases will be employed early in the development stages. Lightweight models will be used to ensure that training and implementation are fast and manageable.

9.3 Resource Requirement

The hardware requirement includes personal computers for development and coding. The software stack comprises Visual Studio Code, Python, Node.js or Django, React.js or Flutter, Firebase, and GitHub.

Considering the defined scope, tools, and timeline, this project is feasible within the academic schedule.

10 Application Areas

RupiSafe holds significant real-world value in the personal finance and fintech domain, serving a variety of user groups. Students can use it to manage their monthly allowances, particularly if they live independently. Employees can benefit from RupiSafe by planning and tracking their expenses between paydays, ensuring better financial control. Families can leverage the application to manage household budgets effectively and navigate financial crises with greater ease.

11 Expertise of Team Members

Contributor	Thesis	Backend & Frontend
Fabiha Adnan	Literature review, writing technical depth (algorithms, models), user interaction & usability study write-up	Frontend, Backend architecture setup
Minal Tahir	Methodology & Proposed System, mathematical definition and formulation, System design & architecture diagrams, Survey design and analysis	Backend, Basic backend-frontend integration support
Both	Final report drafting and editing, Problem statement, objectives	Joint implementation of ML/NLP Models

Table 11.1: Expertise of Team Members

12 Milestones

Milestone	Description	Timeline
Problem Identification and Research	Identify target users, analyze problems and gather requirements.	Week 1 – Week 4
System & UI design	Design system architecture, database schema and user interface mockups	Week 5 – Week 7
AI Model Development	Build and train models	Week 8 – Week 11
Core Application Development	Implement backend, database integration and front end dashboard	Week 12 – Week 16
Testing & Debugging	Conduct functionality, performance and user testing. Fix any bugs and optimize AI modules	Week 17 – Week 20
Final Report and Submission	Finalize documentation, prepare demo and submit completed project	Week 21 – Week 22

Table 12.1: Milestones

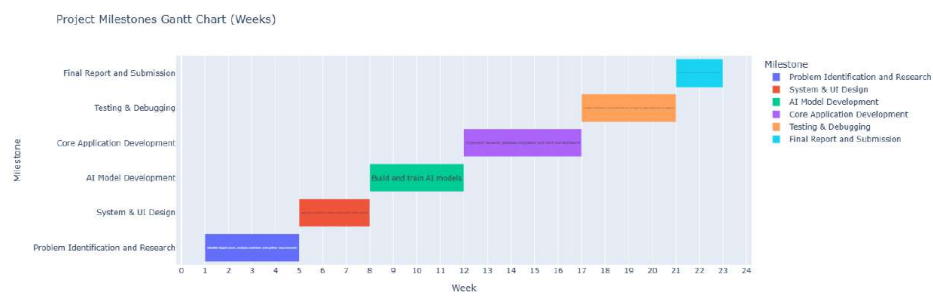


Figure 12.1: Milestone Gantt Chart

13 Expected Outcomes

At the end of the project, we expect to deliver a fully functional budgeting and survival application. A unique Survival Mode algorithm will help users navigate financial crises by optimizing daily allowances and providing practical survival tips. The system will feature AI-driven categorization of expenses and personalized financial advice tailored to individual spending patterns. Monthly reports and what-if simulations will allow users to plan smarter and make informed financial decisions. Overall, the project aims to provide a localized, practical solution for Pakistani users, which can later be scaled to other regions.

14 Significance of Project

This project is important for several reasons. First, it is practically useful, helping students, families, and employees manage their money better. Second, it is innovative, offering a unique Survival Mode that other budgeting apps don't have. Third, it is technically advanced, combining areas like software development, data management, decision support, user experience, and AI. Fourth, it can grow into a full financial assistant with features like bank integration, multiple languages, and an AI chatbot. Finally, it is designed specifically for Pakistani users, making it relevant and more helpful for them.

16 References

- [1] Pakistan Bureau of Statistics, *Pakistan Economic Survey 2023–24*, Government of Pakistan, 2024.
- [2] State Bank of Pakistan, *Household Financial Literacy and Budgeting Practices in Pakistan*, 2023.
- [3] YNAB, *You Need A Budget*, 2024.